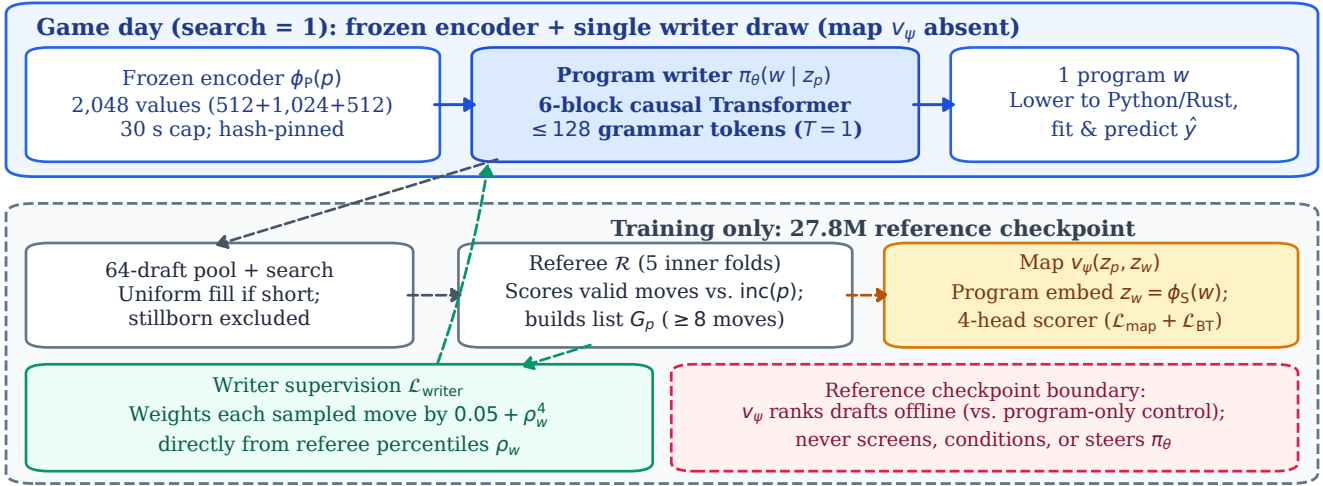
Game day (search = 1): frozen encoder + single writer draw (map $v_\psi$ absent)
Frozen encoder $\phi_P(p)$
2,048 values (512+1,024+512)
30 s cap; hash-pinned
Program writer $\pi_\theta(w \mid z_p)$
6-block causal Transformer
≤ 128 grammar tokens ($T = 1$)
1 program $w$
Lower to Python/Rust,
fit & predict $\hat{y}$
Training only: 27.8M reference checkpoint
64-draft pool + search
Uniform fill if short;
stillborn excluded
Referee $\mathcal{R}$ (5 inner folds)
Scores valid moves vs. inc($p$);
builds list $G_p$ ( ≥ 8 moves)
Map $v_\psi(z_p, z_w)$
Program embed $z_w = \phi_S(w)$;
4-head scorer ($\mathcal{L}_{map} + \mathcal{L}_{BT}$)
Writer supervision $\mathcal{L}_{writer}$
Weights each sampled move by $0.05 + \rho_w^4$
directly from referee percentiles $\rho_w$
Reference checkpoint boundary:
$v_\psi$ ranks drafts offline (vs. program-only control);
never screens, conditions, or steers $\pi_\theta$